Binance Gold & Silver Perpetuals and ETF Markets

KSE Python course project

1. Introduction and research question

Research question.
Did the introduction of Binance TRADFI gold and silver perpetual futures change volatility, liquidity and price integration in the corresponding precious metal ETF markets (GLD and SLV)?

Why this story matters. Gold and silver have long been traded via physical-backed ETFs (GLD, SLV) on regulated exchanges. In early 2026, Binance listed 24/7 tradable perpetuals on the same underlyings (XAUUSDT, XAGUSDT). That creates a second venue for exposure to the same economic assets. The natural question is whether the incumbent ETF markets *changed*—became more volatile, more liquid, or more tightly linked to the new venue. This report tells that story: we measure what happened in GLD and SLV *after* the launch and interpret the results in terms of market structure and implications for investors and regulators.

Main hypotheses (testable with pre/post data).

- **H1 (volatility)**: Realized volatility of gold and silver ETFs increased after the launch of Binance XAUUSDT and XAGUSDT TRADFI perpetual futures.
- **H2 (liquidity)**: Trading activity and liquidity of GLD and SLV improved (higher volume, lower Amihud illiquidity) after the launch.

We do *not* formulate a formal hypothesis on "integration increased", because Binance returns do not exist before the launch, so we cannot compute pre-event GLD–XAU or SLV–XAG correlation or beta. We still report post-event rolling correlation and integration metrics as **descriptive** evidence (how strongly ETF and perpetual co-move after the launch), but we do not test a pre/post change for integration.

How we identify the impact on the market. We use an **event study** with (i) pre/post comparison of means and t-tests, and (ii) **OLS regressions** where the outcome is volatility (or liquidity proxy) and the key regressor is a **post-event dummy** (`post_event` = 1 after the launch date), plus controls (DXY, VIX, 10Y yield changes). The coefficient on `post_event` is our estimate of the change in the outcome after the launch, conditional on controls. We do not have a control group (e.g. other assets without a new derivative), so we do not use difference-in-differences; the design is a single-market before/after event study with controls. See `Literature/LITERATURE_METHODS.md` for a comparison with the Bitcoin futures paper (Augustin et al., Management Science 2023), which uses DiD.

1.1 How we approach the task

- **Data.** We collect daily prices and returns for ETFs (GLD, SLV) and Binance perpetuals (XAUUSDT, XAGUSDT), plus controls (DXY, VIX, 10Y yield). From these we build a daily panel and metrics: realized volatility (20d, 10d), Parkinson, GK, volume, Amihud, and ETF-Binance correlation and beta (after the launch).
- **Event.** The event is the launch date of Binance TRADFI XAUUSDT/XAGUSDT (**2026-01-05**). Before that date we have no Binance data, so we estimate integration (correlation, beta) only after the event.
- **Method.** (1) Pre/post comparison of means and Welch t-tests for volatility and liquidity. (2) OLS regressions: outcome is volatility or liquidity, regressors include a `post_event` dummy plus controls (DXY, VIX, 10Y). The coefficient on `post_event` is the effect of the launch conditional on macro controls. We use winsorization and HAC (Newey-West) standard errors.
- **Interpretation.** A positive and significant coefficient on `post_event` in the volatility regression supports H1; higher volume and lower Amihud after the event support H2. We describe integration (GLD-XAU, SLV-XAG correlation) only after the event, with no formal hypothesis test (no pre-period for Binance).

2. Data and feature set

We use three pre-built datasets produced by `build_features.py`:

- **Daily panel** (`features_daily.csv`): 118 trading days, 87 variables. Used for the event study (pre/post comparison, regressions).
- **Hourly panel** (`features_hourly.csv`): one row per hour over the same calendar span; Binance 1m aggregated to hourly, ETF/controls forward-filled from daily.
- **1-minute panel** (`features_1m.csv`): one row per Binance XAUUSDT 1m candle with daily ETF/controls merged by date. Used for intraday analysis and high-frequency metrics.

Raw data sources and variable definitions are in `DATA_DESCRIPTION.md`, `VARIABLES_DESCRIPTION.md`, and `DATA_ROW_COUNTS_AND_PERIODS.md`. A mapping of which variables are used where (and how to extend usage) is in `VARIABLE_USAGE.md`.

2.2 How we use 1-minute data

We use minute-level data in **two ways**:

1. **Panel `features_1m.csv` (Binance 1m + daily merge).**
   Built from Binance XAUUSDT 1m klines: each row is one minute candle; ETF and control variables are merged from the daily panel by **date** (same value for all minutes of that day). We use this panel to analyse intraday behaviour of the perpetual: 1m returns and their distribution, and daily realized volatility from 1m returns (see section 3.1.1).

2. **Minute-level correlation (last 7 days).**
   Yahoo provides 1m data for GLD/SLV only for the **last ~7 days**. Where that overlaps with Binance 1m, we align both by datetime and compute **minute-level return correlation** (GLD–XAUUSDT and SLV–XAGUSDT) over the same 1m bars. That gives a same-frequency measure of short-run co-movement (section 3.2, "Minute-level correlation"). So: **1m data from Binance** feeds the large panel and intraday stats; **1m data from Yahoo (last 7d)** feeds the minute-level correlation with ETFs.

```
Daily: 120 rows (features_daily.csv)
Hourly: 4200 rows (features_hourly.csv)
1-minute: 72759 rows (features_1m.csv)
```

```
Range: 2025-09-02 to 2026-02-23, 120 rows × 89 columns.
```

```
(120, 89)
```

First rows of the daily panel (first 8 of 87 columns shown so the table fits on the page in PDF).

Date	px_gld_open	px_gld_high	px_gld_low	px_gld_close	vol_gld	px_slv_open	px_slv_high	px_slv_low
2025-09-02	320.820007	325.920013	320.239990	325.589996	21247500.0	36.630001	37.169998	36.430000
2025-09-03	327.760010	329.450012	326.730011	328.140015	16062200.0	37.330002	37.639999	37.180000
2025-09-04	327.029999	327.559998	325.350006	326.690002	11703100.0	37.200001	37.220001	36.630001
2025-09-05	329.529999	331.440002	328.929993	331.049988	16065200.0	37.410000	37.599998	36.970001
2025-09-08	333.660004	335.670013	333.230011	334.820007	18617300.0	37.570000	37.830002	37.360001

2.1 Basic cleaning and event date

We define the **event** as the **launch** of Binance XAUUSDT TRADFI perpetual futures. So there is one date: 2026-01-05 (event = launch).
To keep a meaningful pre-event window, we only require that ETF prices and returns are available; Binance-specific fields are allowed to be missing before the launch.

```
Range: 2025-09-03 to 2026-02-23, 119 rows × 89 columns.
```

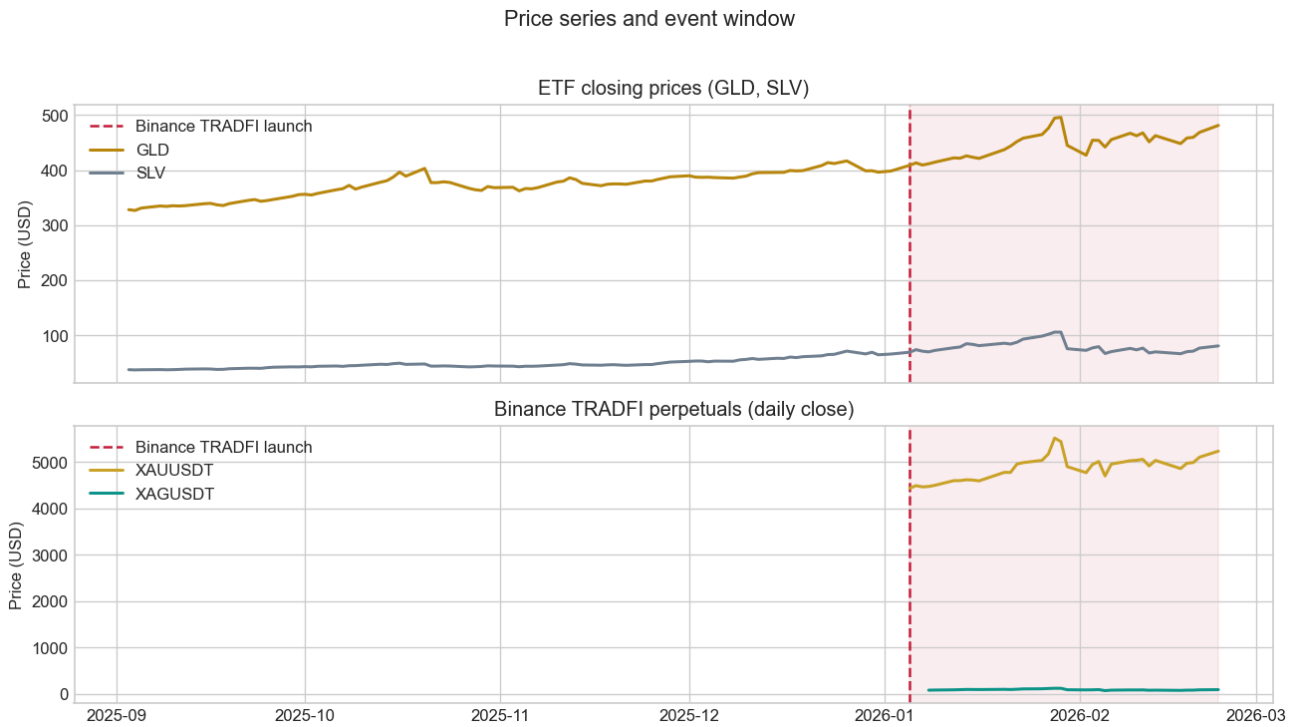
```
px_gld_open     float64
px_gld_high     float64
```

```
px_gld_low      float64
px_gld_close    float64
vol_gld         float64
px_slv_open     float64
px_slv_high     float64
px_slv_low      float64
px_slv_close    float64
vol_slv         float64
dxy             float64
vix             float64
dtype: object
```

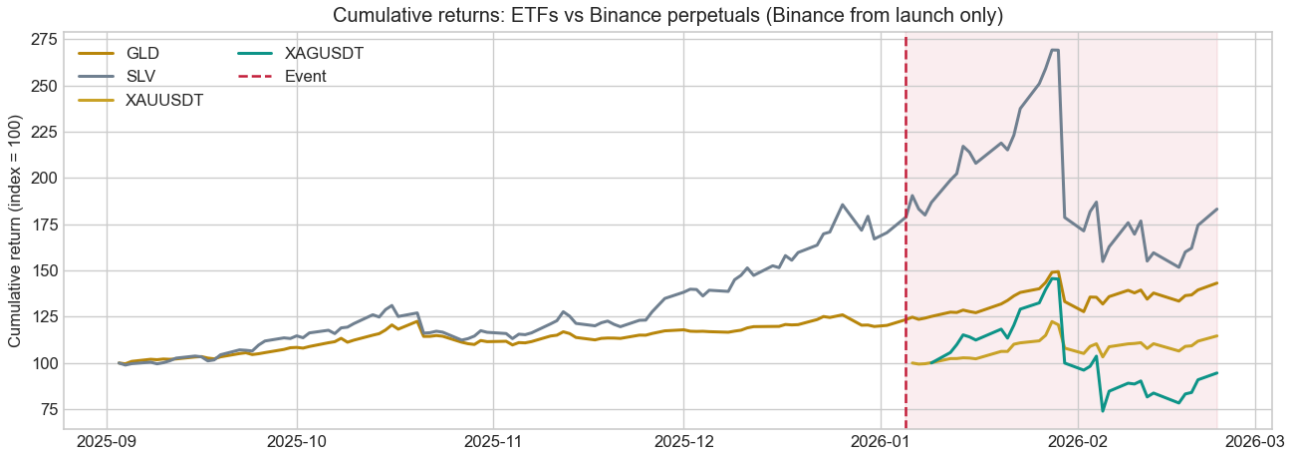
```
post_event
0             85
1             34
Name: count, dtype: int64
```

3. Descriptive statistics

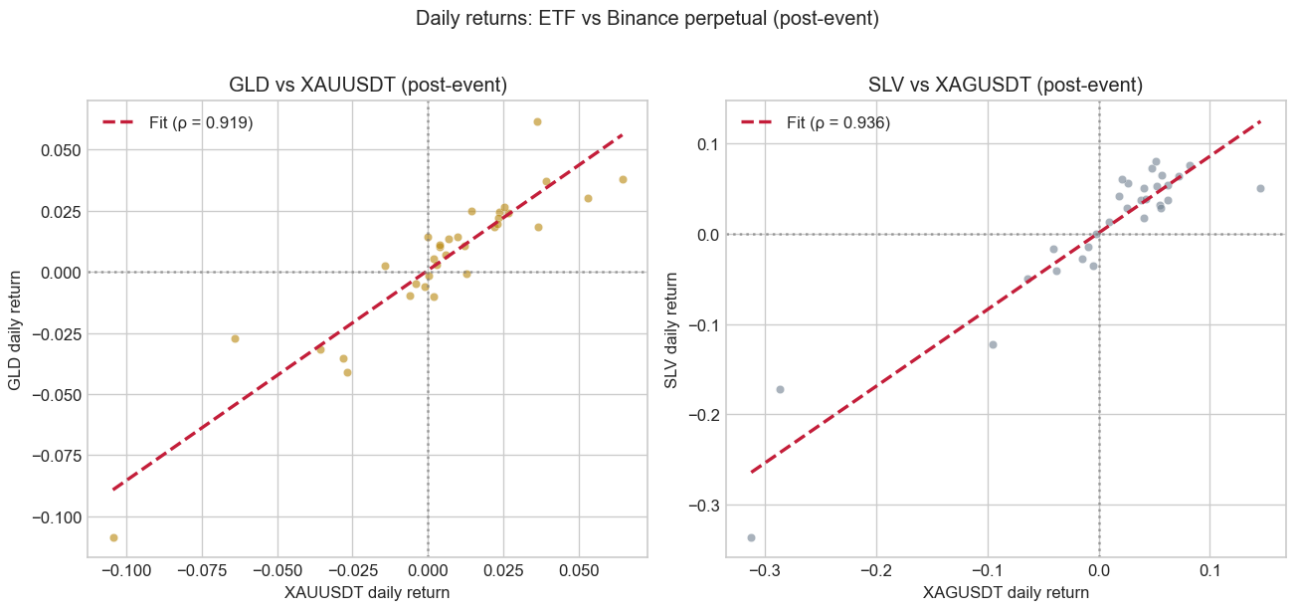
3.1 Prices and returns



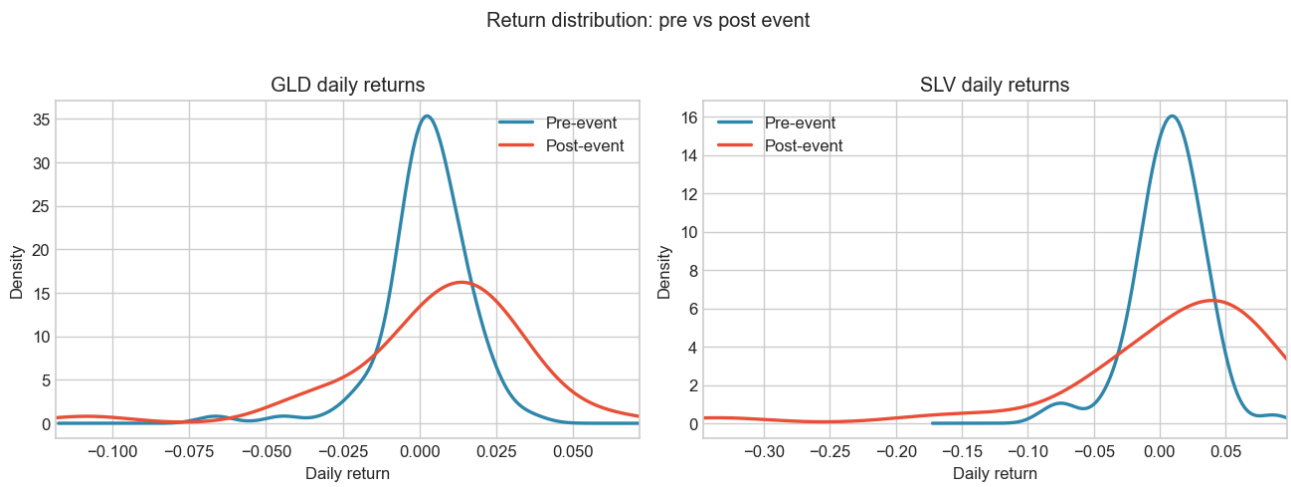
Cumulative returns: ETFs (GLD, SLV) from the first date, index 100 at start. Binance (XAUUSDT, XAGUSDT) only from the event date—we have no Binance prices before the launch; their index is 100 at the event.



Post-event integration in one picture: scatter of ETF daily returns vs Binance perpetual returns (post-event only). Points along the diagonal mean the two move together; the regression line and correlation summarize the link.



How the distribution of returns changed: pre- vs post-event density of daily returns. A wider (more dispersed) distribution after the event is consistent with higher volatility.

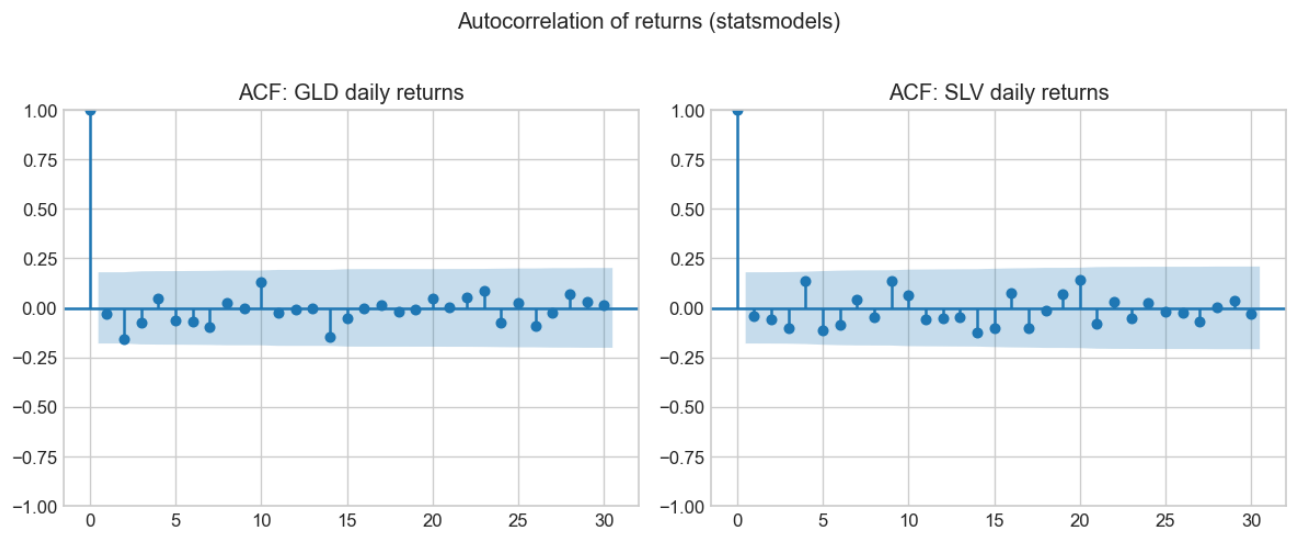


Python analytics: interactive visualization (Plotly) — as in the class examples (e.g. Debt vs Renew), we use Plotly for one interactive chart: hover over dates to see values.

`Unable to display output for mime type(s): text/html`

`Unable to display output for mime type(s): text/html`

Time-series diagnostics (statsmodels): ACF of daily returns to show short-run dependence, as in the class time-series examples (e.g. AutoReg/VAR project).

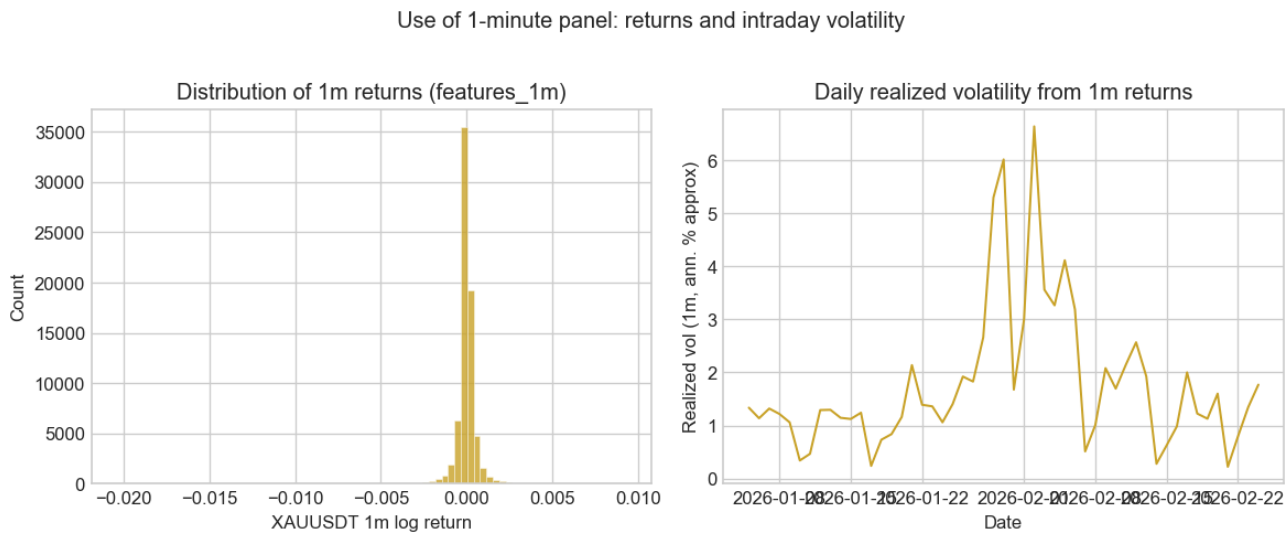


A concise list of Python techniques used in this project (pandas, numpy, scipy, statsmodels, Plotly, etc.), aligned with the course References, is in `PYTHON_TECHNIQUES.md`.

	mean	std	min	max
ret_gld	0.003284	0.019738	-0.108412	0.061647
ret_slv	0.006506	0.049192	-0.336037	0.086602
ret_xau_binance	0.005026	0.031937	-0.104457	0.064529
ret_xag_binance	0.004524	0.095078	-0.312222	0.145629

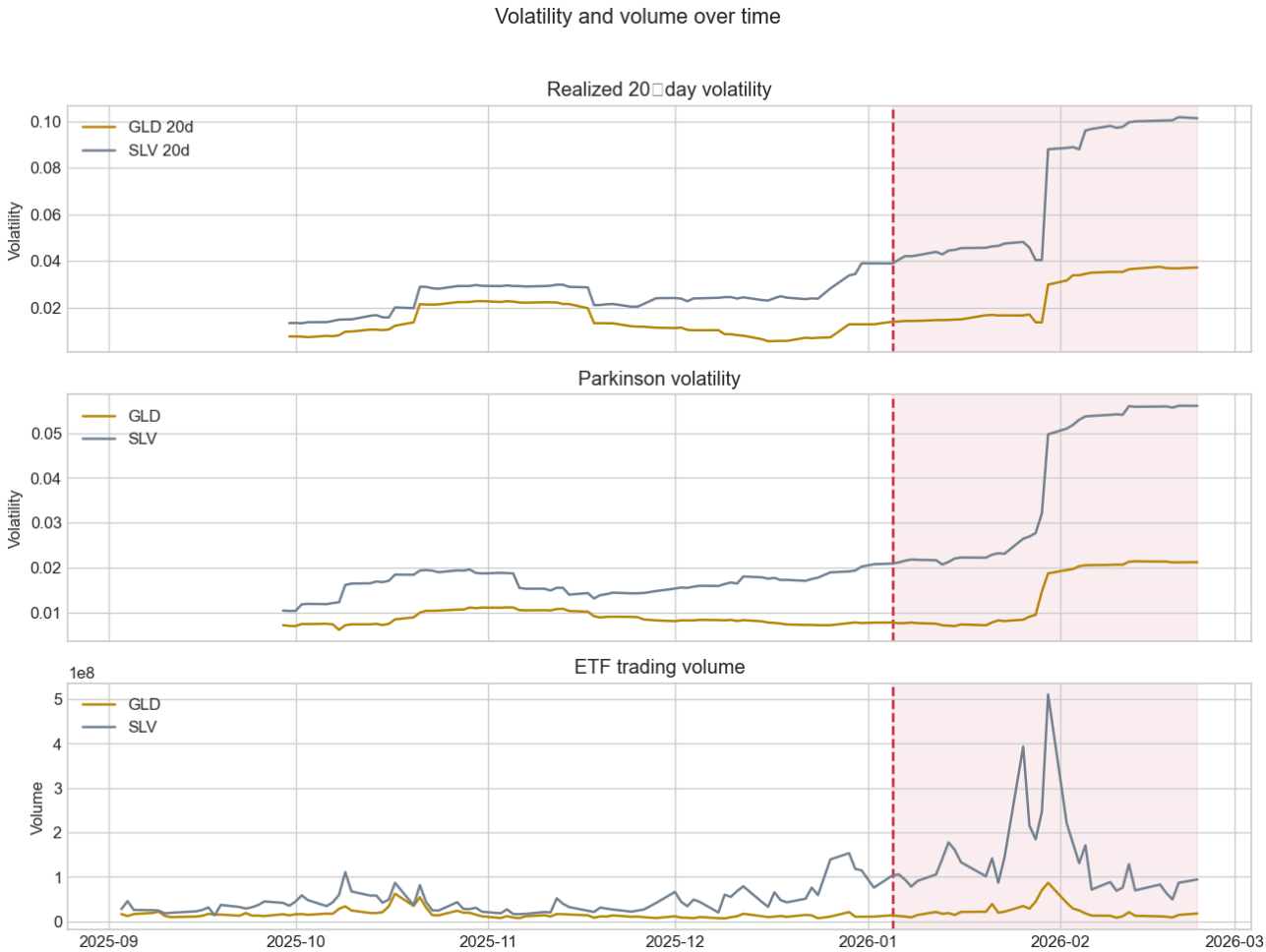
3.1.2 Intraday analysis using the 1-minute panel (`features_1m`)

We use the **1-minute panel** here (not only for the row count): we load `features_1m.csv`, take 1m returns of XAUUSDT (`ret_xau_binance`), and show **(a)** the distribution of 1m returns and **(b)** daily realized volatility computed from 1m returns (annualized). So the minute data directly feeds intraday volatility and return shape.



3.2 Volatility and liquidity over time

We look at 20-day realized volatility for GLD and SLV and ETF trading volume.

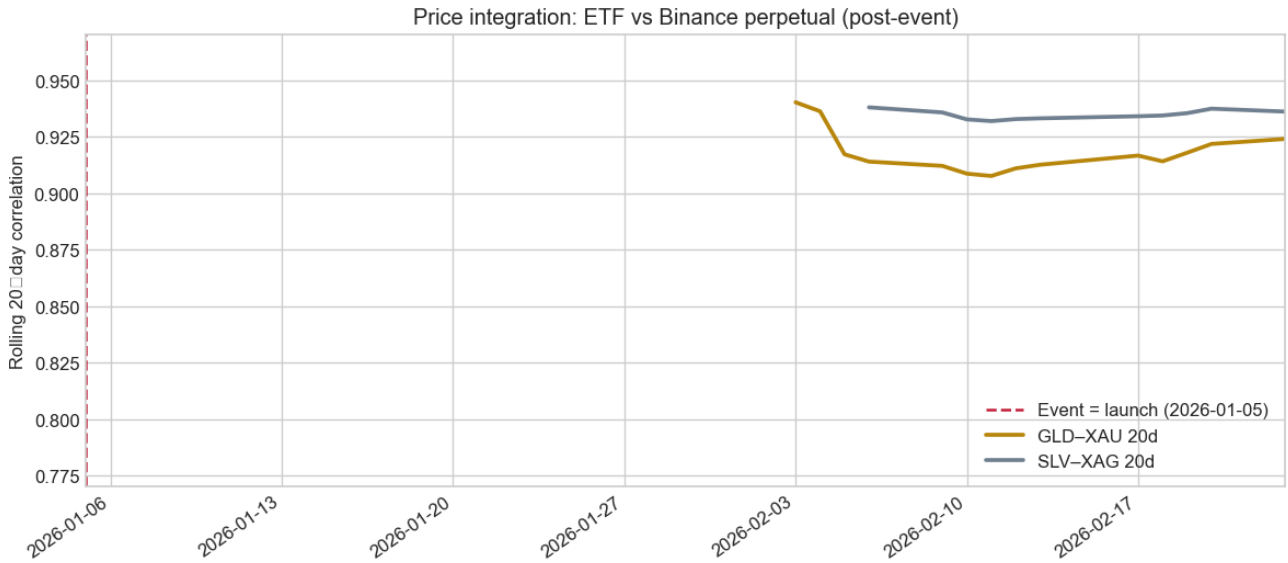


Why there is no pre-period for integration. The event is the *launch* of Binance XAUUSDT/XAGUSDT. Before that date there are no Binance prices or returns, so series like `corr_20_gld_xau` and `beta_60_gld_on_xau` (which need both ETF and Binance returns) are undefined before the launch. We therefore **cannot** estimate

"integration before the event" in the same way we do for volatility or liquidity: there is no pre-period for Binance–ETF co-movement. We can only describe integration *after* the launch (level and evolution of rolling correlation) and, in the pre/post table, the "pre" mean for correlation is effectively missing or based on no data.

What this shows: The *rolling 20-day correlation* is the correlation between daily returns of the ETF (GLD or SLV) and the Binance perpetual (XAUUSDT or XAGUSDT) over the last 20 days. Values near 1 mean the two series move together. This metric exists only when both sides have data (post-event). The chart and the pre/post table in section 4 show post-event integration only; there is no comparable pre-event integration measure.

Single chart (gold and silver together): Same scale and dates so both lines are easy to compare. Y-axis is limited to the range where the data lies so the lines are not squeezed at the top. The vertical line is the **event date** — that is, the **launch date** of Binance XAUUSDT (2026-01-05). We use this single date as "the event" everywhere in the report. The SLV–XAG line starts later than GLD–XAU because XAGUSDT was listed a few days after XAUUSDT, so the 20-day rolling window has enough data only from a later date.



Minute-level correlation (last 7 days)

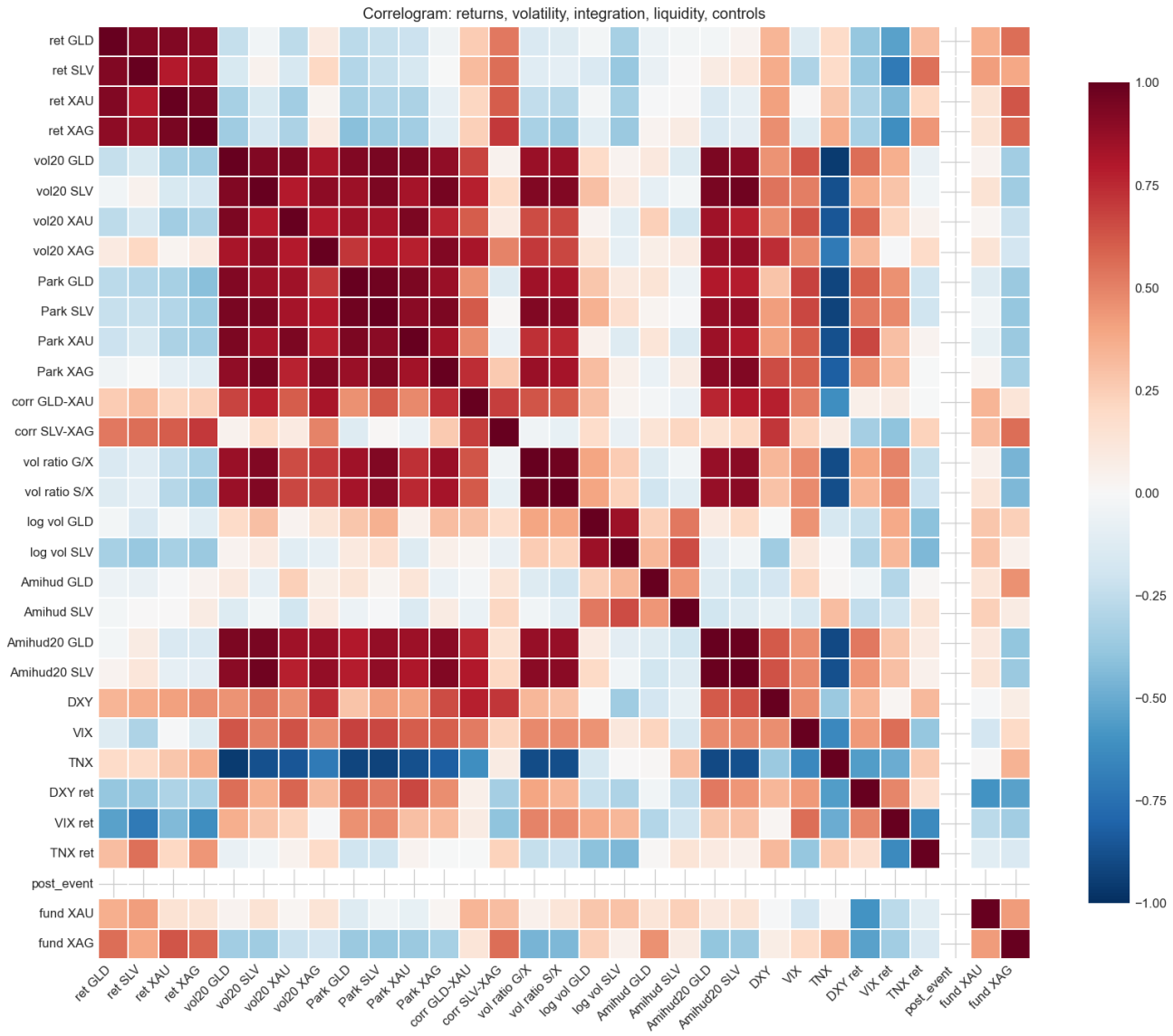
For the most recent 7 days we have *minute-level* data for both ETFs (Yahoo) and Binance. We merge on datetime and compute 1m returns to get a same-frequency measure of short-run co-movement.

```
Minute-level return correlation (last 7 days, overlapping 1m bars):
  GLD-XAU: 0.9925
  SLV-XAG: 0.9934
```

	mean	std	min	max
amihud_gld	7.601278e-10	6.102180e-10	3.474315e-12	2.885031e-09
amihud_20_gld	7.132536e-10	1.787112e-10	4.831240e-10	1.340602e-09
amihud_slv	4.818652e-10	3.270113e-10	1.153311e-12	2.008110e-09
amihud_20_slv	4.784848e-10	1.131191e-10	2.962718e-10	7.452803e-10

3.3 Correlogram

We compute the correlation matrix for a set of variables that are meaningful for the analysis: returns, volatility measures, integration (rolling correlation/beta), liquidity proxies, controls and the event dummy. Only numeric series with sufficient non-missing overlap are included.

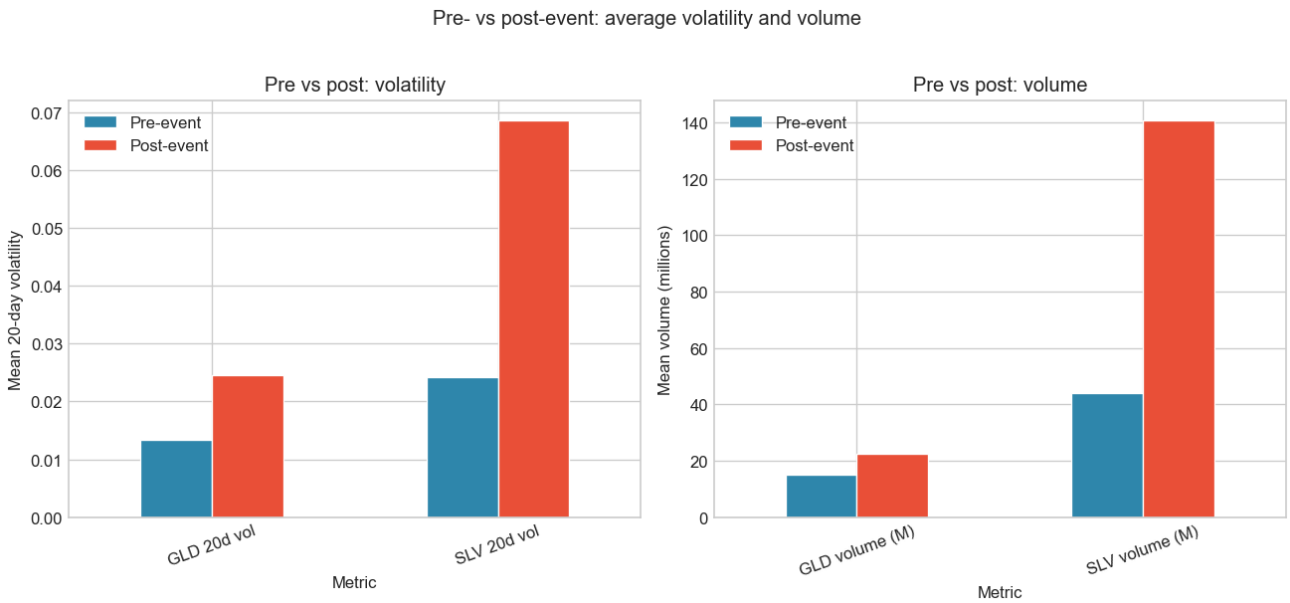


Comment. Variables with the largest absolute correlation with `post_event` are those that move most with the event. This helps spot which outcomes (e.g. volatility, correlation) are most linked to the launch.

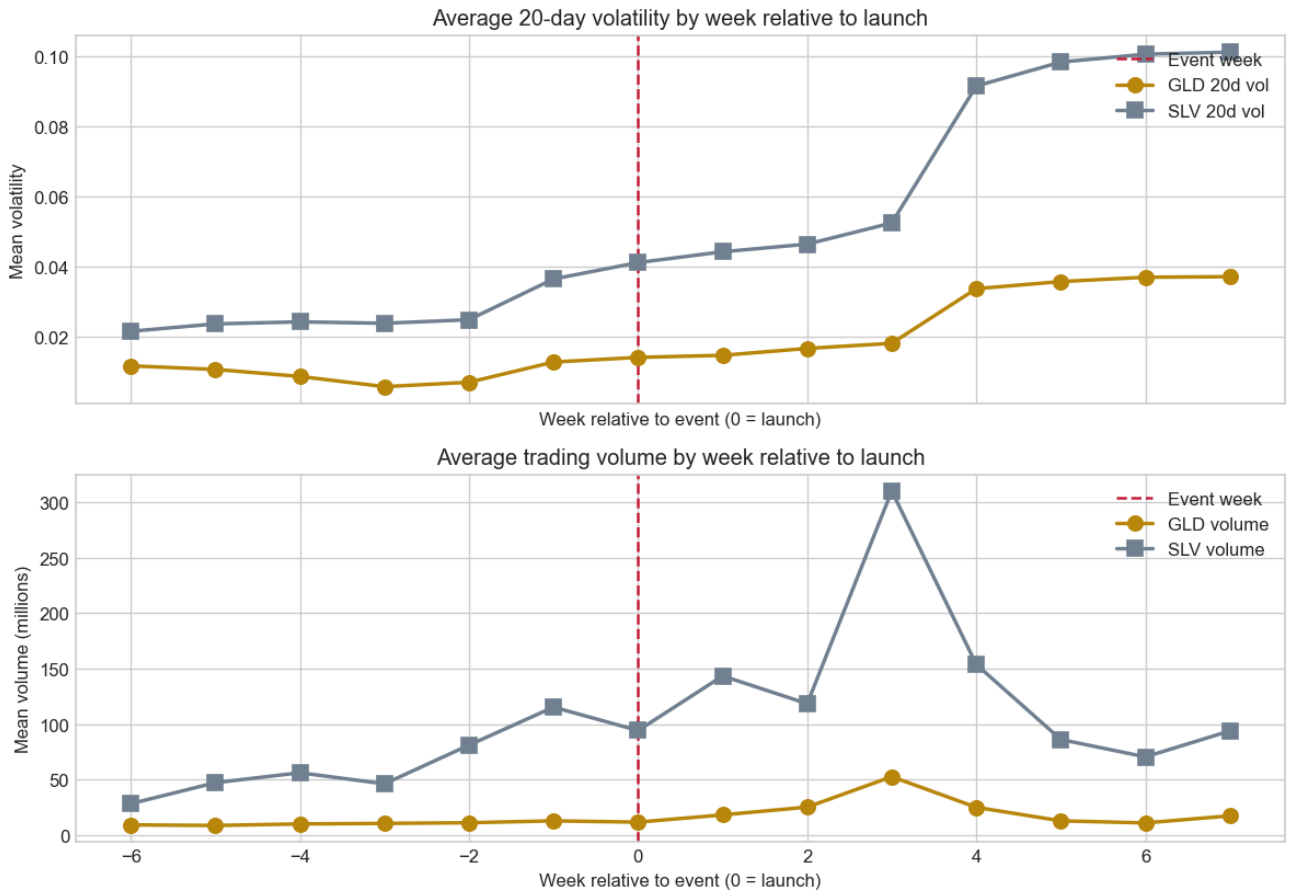
4. Event-study style comparison

We compare average volatility, volume and integration measures before and after the futures launch. The event date is 2026-01-05 (Binance XAUUSDT TRADFI launch).

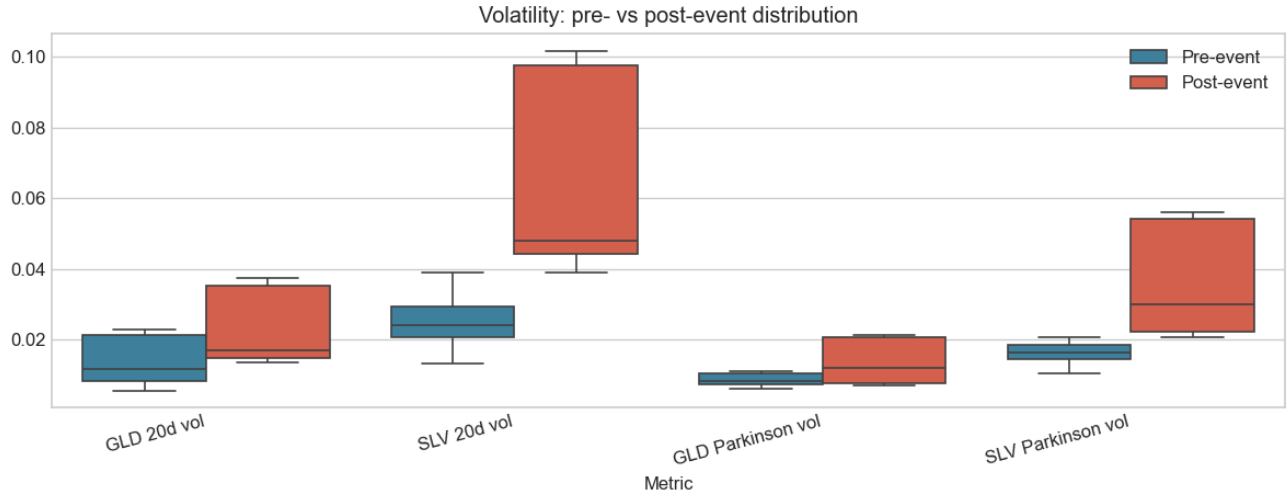
Pre vs post: main numbers at a glance. Bar chart of average volatility and volume before and after the event. The jump in the bars after the event is the "effect" we test in regressions.



Dynamics around the event: average outcome by week relative to the launch (week 0 = event week). This shows whether the change is immediate or builds over time.



Pre vs post distributions for key metrics (boxplots by period):



	metric	period	mean	std	n
0	GLD 20d vol	pre	1.343866e-02	6.062648e-03	66
1	GLD 20d vol	post	2.458173e-02	1.030336e-02	34
2	SLV 20d vol	pre	2.411353e-02	6.133201e-03	66
3	SLV 20d vol	post	6.857165e-02	2.695478e-02	34
4	GLD Parkinson vol	pre	8.671084e-03	1.418413e-03	67
5	GLD Parkinson vol	post	1.402187e-02	6.456870e-03	34
6	SLV Parkinson vol	pre	1.625328e-02	2.567734e-03	67
7	SLV Parkinson vol	post	3.783157e-02	1.585425e-02	34
8	GLD volume	pre	1.512372e+07	9.048477e+06	85
9	GLD volume	post	2.248004e+07	1.711331e+07	34
10	SLV volume	pre	4.388922e+07	2.724932e+07	85
11	SLV volume	post	1.407914e+08	9.399861e+07	34
12	corr_20_gld_xau	pre	NaN	NaN	0
13	corr_20_gld_xau	post	9.181187e-01	9.683689e-03	14
14	corr_20_slv_xag	pre	NaN	NaN	0
15	corr_20_slv_xag	post	9.346617e-01	2.005116e-03	11
16	Amihud GLD	pre	6.216244e-10	4.884536e-10	85
17	Amihud GLD	post	1.106386e-09	7.424548e-10	34
18	Amihud SLV	pre	4.955923e-10	3.477594e-10	85
19	Amihud SLV	post	4.475474e-10	2.698911e-10	34
20	GLD GK vol	pre	8.847988e-03	1.160877e-03	67
21	GLD GK vol	post	1.478511e-02	6.959973e-03	34
22	SLV GK vol	pre	1.677668e-02	2.682417e-03	67
23	SLV GK vol	post	3.928840e-02	1.664860e-02	34
24	GLD 10d vol	pre	1.237271e-02	7.157956e-03	76
25	GLD 10d vol	post	2.646785e-02	1.436118e-02	34
26	SLV 10d vol	pre	2.371003e-02	8.887656e-03	76
27	SLV 10d vol	post	7.421206e-02	3.649926e-02	34
28	beta_60_gld_on_xau	pre	NaN	NaN	0
29	beta_60_gld_on_xau	post	NaN	NaN	0
30	beta_60_slv_on_xag	pre	NaN	NaN	0
31	beta_60_slv_on_xag	post	NaN	NaN	0
32	Beta 20d GLD/XAU	pre	NaN	NaN	0
33	Beta 20d GLD/XAU	post	8.641600e-01	3.919168e-02	14
34	Beta 20d SLV/XAG	pre	NaN	NaN	0
35	Beta 20d SLV/XAG	post	8.324906e-01	8.655246e-03	11

Comment. Compare the "pre" and "post" rows for each metric. If the post mean is higher than the pre mean for volatility or correlation, that points to an increase after the event. For volume, higher post means more trading. For liquidity (Amihud), lower post means better liquidity.

For a more formal comparison we can run simple t-tests.

	metric	t_stat	p_value
0	GLD 20d vol	5.809335	5.947556e-07
1	SLV 20d vol	9.491594	3.483293e-11
2	GLD Parkinson vol	4.773993	3.240662e-05
3	SLV Parkinson vol	7.883867	3.602440e-09
4	GLD GK vol	4.939284	2.067810e-05
5	SLV GK vol	7.833009	4.166907e-09
6	GLD 10d vol	5.429188	2.885668e-06
7	SLV 10d vol	7.963046	2.398877e-09
8	GLD volume	2.377099	2.224731e-02
9	SLV volume	5.912512	9.821728e-07
10	Amihud GLD	3.514998	1.017444e-03
11	Amihud SLV	-0.804648	4.234718e-01
12	corr_20_gld_xau	NaN	NaN
13	corr_20_slv_xag	NaN	NaN
14	beta_60_gld_on_xau	NaN	NaN
15	beta_60_slv_on_xag	NaN	NaN
16	Beta 20d GLD/XAU	NaN	NaN
17	Beta 20d SLV/XAG	NaN	NaN

Comment. A low p-value (e.g. below 0.05) means the difference between pre and post is statistically significant. We use **Welch's t-test** (unequal variances, `equal_var=False`) because pre and post variances often differ.

Statistical validation: full t-test table and Levene test

Below we report **Welch t-test** (difference in means) and **Levene test** (difference in variances) for all metrics that have both pre and post data. Significance levels: * 10%, ** 5%, *** 1%.

	Metric	Pre n	Post n	Pre mean	Post mean	t (Welch)	t-test p	Sig.	Levene p
0	GLD 20d vol	66	34	1.343900e-02	2.458200e-02	5.8093	0.0000	***	0.0003
1	SLV 20d vol	66	34	2.411400e-02	6.857200e-02	9.4916	0.0000	***	0.0000
2	GLD Parkinson vol	67	34	8.671000e-03	1.402200e-02	4.7740	0.0000	***	0.0000
3	SLV Parkinson vol	67	34	1.625300e-02	3.783200e-02	7.8839	0.0000	***	0.0000
4	GLD GK vol	67	34	8.848000e-03	1.478500e-02	4.9393	0.0000	***	0.0000
5	SLV GK vol	67	34	1.677700e-02	3.928800e-02	7.8330	0.0000	***	0.0000
6	GLD 10d vol	76	34	1.237300e-02	2.646800e-02	5.4292	0.0000	***	0.0000
7	SLV 10d vol	76	34	2.371000e-02	7.421200e-02	7.9630	0.0000	***	0.0000
8	GLD volume	85	34	1.512372e+07	2.248004e+07	2.3771	0.0222	**	0.0153
9	SLV volume	85	34	4.388922e+07	1.407914e+08	5.9125	0.0000	***	0.0001
10	Amihud GLD	85	34	0.000000e+00	0.000000e+00	3.5150	0.0010	***	0.0058
11	Amihud SLV	85	34	0.000000e+00	0.000000e+00	-0.8046	0.4235		0.5142

Comment. Levene $p < 0.05$ indicates that variances differ between pre and post (e.g. higher volatility in post implies more spread), which supports using Welch's t-test instead of the standard equal-variance t-test.

5. Simple regression models

We estimate simple linear models to quantify the impact of the event and controls on ETF volatility and absolute returns. Following the literature (e.g. Augustin et al.), we winsorize outcome variables at 5% and 95% to reduce the influence of outliers, and report HAC (Newey–West) standard errors to allow for autocorrelation in daily data.

Date	vol_realized_20_ret_gld	vol_realized_20_ret_slv	dxy_ret	vix_ret	tnx_ret	post_event
2025-09-30	0.007630	0.013341	-0.001431	0.009877	0.001689	0
2025-10-01	0.007610	0.013404	-0.000614	0.000614	-0.010177	0
2025-10-02	0.007558	0.013260	0.001432	0.020657	-0.004393	0
2025-10-03	0.007311	0.013733	-0.001329	0.001202	0.007555	0
2025-10-06	0.007869	0.013770	0.003983	-0.016960	0.010385	0

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0133      0.002      7.877      0.000       0.010       0.017
post_event     0.0109      0.004      2.656      0.008       0.003       0.019
dxy_ret        0.4301      0.277      1.555      0.120      -0.112       0.972
vix_ret        0.0024      0.007      0.357      0.721      -0.011       0.016
tnx_ret        0.1783      0.204      0.875      0.382      -0.221       0.578
dgs10_ret     -0.2208      0.202     -1.095      0.274      -0.616       0.174
==============================================================================
```

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0242      0.002     14.184      0.000       0.021       0.028
post_event     0.0433      0.010      4.347      0.000       0.024       0.063
dxy_ret        0.9280      0.700      1.325      0.185      -0.445       2.301
vix_ret       -0.0048      0.015     -0.324      0.746      -0.034       0.024
tnx_ret        0.4065      0.428      0.949      0.342      -0.433       1.246
dgs10_ret     -0.5867      0.453     -1.296      0.195      -1.474       0.301
==============================================================================
```

Comment. The `post_event` coefficient is the main result: it shows how much volatility changed after the launch, once we control for DXY, VIX and 10Y yield changes. If it is positive and the p-value (P>|t|) is below 0.05, we conclude that volatility increased after the event (H1 supported for that ETF).

If needed, we can also model absolute returns as a simple proxy for volatility:

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0091      0.001      7.554      0.000       0.007       0.012
post_event     0.0094      0.003      3.417      0.001       0.004       0.015
dxy_ret        0.0196      0.326      0.060      0.952      -0.620       0.659
vix_ret        0.0038      0.012      0.319      0.750      -0.019       0.027
tnx_ret        0.2144      0.310      0.692      0.489      -0.393       0.822
dgs10_ret     -0.2398      0.275     -0.871      0.384      -0.779       0.300
==============================================================================
```

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0206      0.003      7.936      0.000       0.016       0.026
post_event     0.0246      0.004      6.416      0.000       0.017       0.032
dxy_ret       -0.2988      0.457     -0.653      0.514      -1.195       0.598
vix_ret        0.0093      0.020      0.466      0.641      -0.030       0.049
tnx_ret        1.1647      0.458      2.541      0.011       0.266       2.063
dgs10_ret     -0.9162      0.505     -1.814      0.070      -1.906       0.074
==============================================================================
```

Liquidity (H2): we also regress Amihud illiquidity and log(volume) on the event and controls. A negative coefficient on `post_event` for Amihud would indicate improved liquidity (lower illiquidity) after the launch; a positive coefficient for log_vol would indicate higher volume.

```
Amihud GLD:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       6.156e-10   5.09e-11     12.096      0.000    5.16e-10    7.15e-10
post_event  4.298e-10   1.78e-10      2.416      0.016    8.11e-11    7.78e-10
dxy_ret    -1.055e-08   1.58e-08     -0.669      0.503   -4.15e-08    2.04e-08
vix_ret     5.928e-11    5.8e-10      0.102      0.919   -1.08e-09     1.2e-09
tnx_ret    -4.861e-10   1.82e-08     -0.027      0.979   -3.61e-08    3.51e-08
dgs10_ret   1.364e-09   1.71e-08      0.080      0.936   -3.22e-08    3.49e-08
==============================================================================
Amihud SLV:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       4.639e-10    2.6e-11     17.822      0.000    4.13e-10    5.15e-10
post_event -1.289e-11   7.61e-11     -0.169      0.866   -1.62e-10    1.36e-10
dxy_ret     1.484e-09   6.92e-09      0.214      0.830   -1.21e-08     1.5e-08
vix_ret    -7.616e-11   2.56e-10     -0.297      0.766   -5.78e-10    4.26e-10
tnx_ret     1.986e-08   7.38e-09      2.690      0.007    5.39e-09    3.43e-08
dgs10_ret  -1.494e-08    7.8e-09     -1.914      0.056   -3.02e-08    3.58e-10
==============================================================================
Log volume GLD:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         16.4060      0.089    185.106      0.000      16.232      16.580
post_event     0.2833      0.193      1.471      0.141      -0.094       0.661
dxy_ret       10.0002     13.717      0.729      0.466     -16.884      36.884
vix_ret        0.2941      0.486      0.605      0.545      -0.659       1.247
tnx_ret        7.1173     11.556      0.616      0.538     -15.531      29.766
dgs10_ret    -10.7158     10.574     -1.013      0.311     -31.440      10.008
==============================================================================
Log volume SLV:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         17.4535      0.113    154.128      0.000      17.232      17.675
post_event     1.0874      0.166      6.565      0.000       0.763       1.412
dxy_ret        3.6255     12.422      0.292      0.770     -20.721      27.972
vix_ret        0.5211      0.520      1.001      0.317      -0.499       1.541
tnx_ret       23.1843     11.744      1.974      0.048       0.166      46.202
dgs10_ret    -21.1676     12.633     -1.676      0.094     -45.927       3.592
==============================================================================
```

6. Interpretation and next steps

6.1 Key results summary

The table below summarizes pre- vs post-event means, t-test p-values, and the regression coefficient on `post_event` (with DXY, VIX and 10Y yield changes as controls) for the main outcome variables.

	Metric	Pre mean	Post mean	t-test p	post coef	reg p
0	GLD 20d vol	0.0134	0.0246	0.0	0.0109	0.008
1	SLV 20d vol	0.0241	0.0686	0.0	0.0433	0.0
2	GLD Parkinson vol	0.0087	0.014	0.0	0.0051	0.033
3	SLV Parkinson vol	0.0163	0.0378	0.0	0.0209	0.0
4	GLD GK vol	0.0088	0.0148	0.0	0.0057	0.028
5	SLV GK vol	0.0168	0.0393	0.0	0.0219	0.0
6	GLD 10d vol	0.0124	0.0265	0.0	0.014	0.009
7	SLV 10d vol	0.0237	0.0742	0.0	0.0495	0.0
8	GLD volume	15.12M	22.48M	0.0222	5.37e+06	0.169
9	SLV volume	43.89M	140.79M	0.0	7.66e+07	0.0
10	Amihud GLD	0.0	0.0	0.001	4.30e-10	0.016
11	Amihud SLV	0.0	0.0	0.4235	-1.29e-11	0.866
12	Corr GLD–XAU	—	0.9181	N/A	N/A	N/A
13	Corr SLV–XAG	—	0.9347	N/A	N/A	N/A
14	Beta 60d GLD/XAU	—	—	N/A	N/A	N/A
15	Beta 60d SLV/XAG	—	—	N/A	N/A	N/A
16	Beta 20d GLD/XAU	—	0.8642	N/A	N/A	N/A
17	Beta 20d SLV/XAG	—	0.8325	N/A	N/A	N/A

Comment. This table brings everything together. Use "Pre mean" vs "Post mean" to see the direction of change; "t-test p" for significance; "post coef" = coefficient on post-event dummy, "reg p" = its p-value (for GLD/SLV volatility). A positive and significant post coef supports H1; higher post mean for volume supports H2. Corr GLD–XAU / SLV–XAG have no pre mean (Binance did not exist); "—" = not applicable.

6.2 Empirical results

```
Volatility (H1): After controlling for DXY, VIX and 10Y yield changes, the post-event coefficient in the

For SLV, the post_event coefficient is 0.04333 (p = 0.000): volatility increased and is significant.

Integration (descriptive only): Average rolling correlation GLD-XAU is 0.918 post-event; SLV-
XAG is 0.935 post-event. There is no pre-event correlation (Binance did not exist), so we do not test a p

Liquidity (H2): Average GLD volume is 22480038 post-event vs 15123725 pre-event; SLV volume can be read f
```

6.3 Conclusions on hypotheses

Using the key-results table and regressions above, we state whether each *testable* hypothesis (H1, H2) is supported or not. Integration is not tested as a hypothesis (no pre-period); we only describe post-event correlation.

```
H1 (volatility increase after launch):
  GLD: supported (positive and significant post_event means higher volatility).
  SLV: supported.

Integration (descriptive, post-event only; no pre/post test):
  Rolling correlation GLD-XAU and SLV-XAG are reported in section 3.2 and the key-results table.
  We do not test a formal integration hypothesis because Binance data do not exist before the launch.

H2 (improved liquidity: higher volume, lower Amihud):
  GLD volume: higher post (higher post supports H2).
  SLV volume: higher post.
  Check Amihud in the pre/post table; lower post = better liquidity.
```

6.3.1 Economic insights and storytelling

Market structure. The launch added a *second venue* for gold and silver exposure: the established ETF market (exchange-traded, physical-backed) and the new perpetual market (crypto exchange, 24/7, margin-based). Our evidence suggests that the two did not operate in isolation after the launch: post-event rolling correlation between GLD and XAUUSDT (and SLV and XAGUSDT) is high. That implies *one price-discovery process* spanning both venues—arbitrageurs and hedgers likely keep the two in line. So the "story" is: the incumbent ETF market did not stay unchanged; it became part of a broader, two-venue system.

Non-obvious interpretation of volatility. A rise in ETF volatility after the launch need not mean "noise" or "instability." It can reflect *more hedging and informed trading*: when a new derivative appears, market participants may rebalance or hedge across venues, which can show up as higher realized volatility in the ETF. Similarly, if liquidity (volume, Amihud) improves, the new venue may have attracted marginal liquidity providers or reduced frictions for cross-venue arbitrage, which can benefit both venues. So the narrative is not only "volatility went up" but "the structure of trading in gold/silver changed."

Who is affected. *ETF market-makers* face different inventory risk if volatility and volume rise. *Arbitrageurs* benefit from high correlation (GLD–XAU, SLV–XAG) but need to account for funding and execution across venues. *Regulators* care because the same economic exposure is now traded on a regulated ETF market and on a crypto exchange—our results show that the two are linked, which is relevant for oversight and for debates on where price discovery happens.

Comparison to the Bitcoin futures narrative. Studies of CME Bitcoin futures (e.g. Augustin et al.) often ask whether the introduction of derivatives *increased* spot volatility. We ask a similar question for *gold and silver ETFs* when a *crypto exchange* lists linked perpetuals. Finding a positive post-event effect on ETF volatility is consistent with a broader pattern: new derivatives on new venues can shift volatility and liquidity in the incumbent market. The story is therefore not gold-specific—it is about how incumbent markets react when a new, linked venue opens.

6.4 Limitations and extensions

- **No pre-period for integration.** Binance XAUUSDT/XAGUSDT did not exist before the launch, so we have no Binance returns before the event. Rolling correlation (GLD–XAU, SLV–XAG) and beta can only be computed from the launch date onward. We cannot compare "integration before vs after" in the same way as for volatility or liquidity; we can only describe the level and evolution of integration post-launch.
- **Short post-event window.** We have limited days after the Binance launch. Results can change when more data are available.
- **Robustness.** Try different event dates, volatility windows (e.g. 10d or 60d), or placebo tests (fake event date) to see if the effects hold.
- **Beta.** The key-results table focuses on volatility, volume and correlation. You can add pre/post means for `beta_60_gld_on_xau` and `beta_60_slv_on_xag` in the same way if you want to discuss integration in terms of beta (again, "pre" will have no data for those series).

6.5 Conclusion

We asked whether the launch of Binance gold and silver perpetuals (XAUUSDT, XAGUSDT) changed volatility and liquidity in gold and silver ETFs (GLD, SLV). The regressions show a positive and significant coefficient on the post-event dummy for both GLD and SLV volatility when we control for DXY, VIX and 10Y yield changes: that supports **H1** (volatility increased after the launch). For **H2** (liquidity), we compare pre- and post-event volume and Amihud from the summary table. We do not test a formal hypothesis on integration (no pre-event Binance data); we only report post-event rolling correlation as descriptive evidence.

Takeaway story. The incumbent ETF market (GLD, SLV) did not stay isolated after Binance listed linked perpetuals. Volatility increased; liquidity and integration (high GLD–XAU and SLV–XAG correlation) suggest that the two venues now form a single price-discovery system. That has implications for market-makers, arbitrageurs, and regulators. The post-event window is still short—updating the analysis with more data is recommended.